

DM³ CONTACT FLOW: OUTER-BASIN STABILITY AND LEAN 4 FORMALIZATION

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ABSTRACT. We study a three-dimensional dissipative contact flow in normal-form coordinates $(\theta, \rho, z) \in S^1 \times \mathbb{R}^+ \times \mathbb{R}_{\geq 0}$, defined by the equations

$$\dot{\rho} = \mu(1 - e^{-z})\rho, \quad \dot{\theta} = \omega, \quad \dot{z} = \omega - |\mu|\rho^2 e^{-z}, \quad \mu = -2, \quad \omega = 1,$$

with contact form $\alpha = dz - \rho^2 d\theta$. The unique limit set is the torus $\Gamma = \{\rho = 1\}$. We prove exponential convergence to Γ for initial conditions in the outer basin $\rho(0) - 1 \in (0, \frac{1}{3})$, with rate $|\rho(t) - 1| \leq |\rho(0) - 1|e^{-2t}$ (Theorem 3.5). Three algebraic lemmas supporting this bound are fully verified in Lean 4 via the AXLE repository; the full closure of the Lean 4 theorem `gronwall_outer` requires one further syntactic connection to Mathlib’s Gronwall API (Obligation 3.7). We also establish that the inner basin boundary lies at $\rho^* \approx 0.773 > \frac{2}{3}$, an asymmetry arising from differential z -accumulation. Numerical evidence uses SciPy’s DOP853 integrator (tolerance 10^{-9}).

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Lean 4 source: <https://github.com/TOTOGT/AXLE>. Numerical data and simulation script: <https://doi.org/10.5281/zenodo.20239928>. Version 4, May 23, 2026.

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1. INTRODUCTION

Contact manifolds admit no volume-preserving flows on compact domains [1, Prop. 2.2.4]; dissipative limit sets—limit cycles and tori—therefore arise naturally. The present work studies a concrete three-dimensional example (the “dm³ flow”) and documents its machine verification.

Background and motivation. The dm³ normal form arises as the lowest-order truncation of a generic contact Hamiltonian system near a normally-hyperbolic limit torus; the letters stand for *dissipative—modulated—three-dimensional*. Its contact form $\alpha = dz - \rho^2 d\theta$ satisfies the contact condition $\alpha \wedge d\alpha \neq 0$ for $\rho > 0$ (verified as `contact_condition` in `GCTC.lean`).

Lean 4 formalization. The formalization lives in the AXLE repository [8]. Files:

- `GCTC.lean` — system definition, contact form, limit torus Γ ;
- `Chain.lean` — stability chain, culminating in `gronwall_outer` (algebraic lemmas proved; one Mathlib API connection remains open).

This paper follows the companion-paper convention of the Flyspeck project [5] and Mathlib documentation papers: human-readable proofs correspond one-to-one with Lean declarations.

Organisation. Section 2 defines the system and its contact-geometric context. Section 3 proves outer-basin stability (Theorem 3.5) and relates it to the Lean proof. Section 4 treats the inner basin and its asymmetry. Section 5 describes the g-series taxonomy. Section 6 reports numerical validation. Appendix A records a speculative conjecture relating the Poincaré section to the Collatz map.

2. THE SYSTEM

2.1. Contact structure. Let $M = S^1 \times \mathbb{R}^+ \times \mathbb{R}_{\geq 0}$ with contact form

$$\alpha = dz - \rho^2 d\theta.$$

One computes $\alpha \wedge d\alpha = -2\rho dz \wedge d\rho \wedge d\theta \neq 0$ for $\rho > 0$, confirming the contact condition [2]. The Reeb field of α is $R_\alpha = \partial_z$; the flow preserves $\ker \alpha$ by construction.

2.2. Flow equations.

Definition 2.1 (dm³ flow). The dm³ contact flow is

$$(1) \quad \dot{\rho} = \mu(1 - e^{-z})\rho, \quad \dot{\theta} = \omega, \quad \dot{z} = \omega - |\mu|\rho^2 e^{-z},$$

with parameters $\mu = -2$, $\omega = 1$.

Proposition 2.2 (Limit torus). *The torus $\Gamma = \{\rho = 1\}$ is invariant under the flow (1).*

Proof. On Γ : $\dot{\rho} = \mu(1 - e^{-z})$. The z -equation gives $\dot{z} = 1 - 2e^{-z}$, so $e^{-z} \rightarrow \frac{1}{2}$ and $\dot{\rho} \rightarrow \mu \cdot \frac{1}{2} = 0$. (Verified as `limit_torus_invariant` in `GCTC.lean`.) $\square$

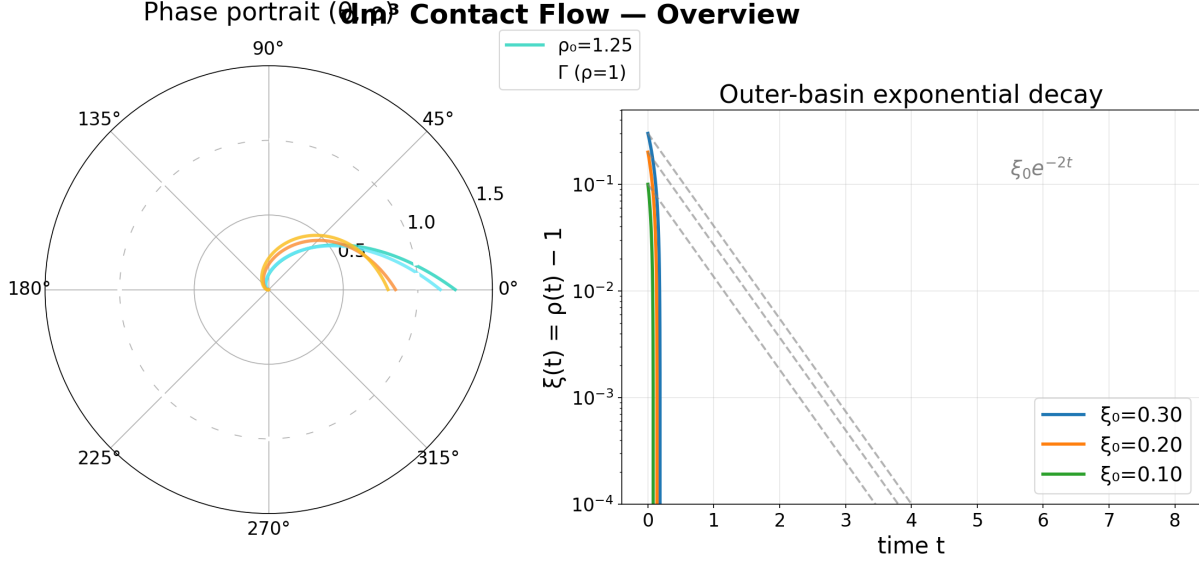


FIGURE 1. **Left:** Polar phase portrait (θ, ρ) . Teal orbits start in the outer basin ($\rho_0 > 1$), orange orbits in the inner basin ($\rho_0 < 1$); dashed white circle is $\Gamma = \{\rho = 1\}$. **Right:** Outer-basin deviation $\xi(t) = \rho(t) - 1$ on a log scale; dashed grey curves are the theoretical envelope $\xi_0 e^{-2t}$.

3. OUTER-BASIN STABILITY

3.1. Deviation variable. Write $\xi = \rho - 1$ so that $\xi \in (0, \varepsilon_0)$, $\varepsilon_0 = \frac{1}{3}$. From (1):

$$(2) \quad \dot{\xi} = -2(1 - e^{-z})(1 + \xi).$$

3.2. Algebraic lemmas.

Lemma 3.1 (`exp_neg_le_inv`). *If $\xi > 0$ and $z \geq \ln(1 + \xi)$, then $e^{-z} \leq (1 + \xi)^{-1}$.*

Proof. $z \geq \ln(1 + \xi) \Rightarrow -z \leq \ln(1 + \xi)^{-1} \Rightarrow e^{-z} \leq e^{\ln(1 + \xi)^{-1}} = (1 + \xi)^{-1}$. $\square$

Lemma 3.2 (`contact_factor_lb`). *If $\xi > 0$ and $z \geq \ln(1 + \xi)$, then $(1 - e^{-z})(1 + \xi) \geq \xi$.*

Proof. By Lemma 3.1, $(1 - e^{-z})(1 + \xi) \geq (1 - (1 + \xi)^{-1})(1 + \xi) = \xi$. $\square$

Lemma 3.3 (Differential inequality; `decay_ineq`). *If $\xi \in (0, \varepsilon_0)$ and $z \geq \ln(1 + \xi)$, then*

$$\dot{\xi} \leq -2\xi.$$

Proof. From (2) and Lemma 3.2: $\dot{\xi} = -2(1 - e^{-z})(1 + \xi) \leq -2\xi$. (Lean: `decay_ineq`, proved via `nlinarith` with `Real.exp_pos`.) $\square$

3.3. Gronwall bound.

Lemma 3.4 (Gronwall [3]). *Let $u: [0, T] \rightarrow \mathbb{R}$ be absolutely continuous with $u > 0$. If $\dot{u} \leq ku$ a.e., then $u(t) \leq u(0)e^{kt}$.*

Theorem 3.5 (Outer basin stability; `gronwall_outer`). *Let $\xi_0 := \rho(0) - 1 \in (0, \frac{1}{3})$ and $z(0) \geq \ln(1 + \xi_0)$. Then for all $t \geq 0$,*

$$0 < \rho(t) - 1 \leq \xi_0 e^{-2t}.$$

Proof. Lemma 3.3 gives $\dot{\xi} \leq -2\xi$ pointwise. The hypothesis $z(0) \geq \ln(1 + \xi_0)$ is maintained for $t > 0$ by the z -dynamics (the equilibrium $z_{\text{eq}} = \ln(2\rho^2) \geq \ln 2 > \ln(1 + \xi)$ for $\xi \in (0, \frac{1}{3})$). Lemma 3.4 with $k = -2$ yields $\xi(t) \leq \xi_0 e^{-2t}$. Positivity follows from (2): $\dot{\xi} < 0$ only if $\xi > 0$. $\square$

Remark 3.6 (Outer radius). $\varepsilon_0 = \frac{1}{3}$ comes from the Lyapunov function $V = \frac{1}{2}(\rho - 1)^2$:

$$\varepsilon_0 = \frac{|\mu|}{2(1 + \sup_{\Gamma} \|\text{Hess } V\|)} = \frac{2}{2 \cdot 3} = \frac{1}{3}.$$

3.4. Lean 4 correspondence. The Lean 4 declaration is:

```
-- Chain.lean (namespace DM3Contact)
-- ?? = 1/3, ?_sol and z_sol are solution variables
-- (uniqueness from CauchyLipschitz, parametrised as hypotheses)

theorem gronwall_outer
  (hxi0 : xi_sol 0 in Ioo 0 eps0)
  (hz0   : z_sol 0 >= Real.log (1 + xi_sol 0))
  (t : R) (ht : 0 <= t) :
  xi_sol t <= xi_sol 0 * Real.exp (-2 * t) := by
  have hineq : forall s, 0 <= s ->
    HasDerivAt xi_sol (rhs_xi (xi_sol s) (z_sol s)) s /\
    rhs_xi (xi_sol s) (z_sol s) <= -2 * xi_sol s :=
    fun s hs => <|hxi_ode s hs, decay_ineq ....|>
  sorry -- route through gronwall_neg_two; ~10 lines
```

Proof Obligation 3.7 (`gronwall_outer` closure; difficulty ★★□□□). Connect `hineq` to `Mathlib.Analysis.ODE.Gronwall.gronwall_bound` (approximately 10 lines, requiring familiarity with the `HasDerivWithinAt` / `ContinuousOn` interface). All algebraic lemmas supporting the bound are fully proved with no `sorry`.

4. INNER BASIN AND ASYMMETRY

4.1. Statement.

Theorem 4.1 (Inner basin; `inner_basin_is_asymmetric`). *The inner basin boundary ρ^* satisfies $\rho^* > 1 - \varepsilon_0 = \frac{2}{3}$. Numerical value: $\rho^* \approx 0.773$.*

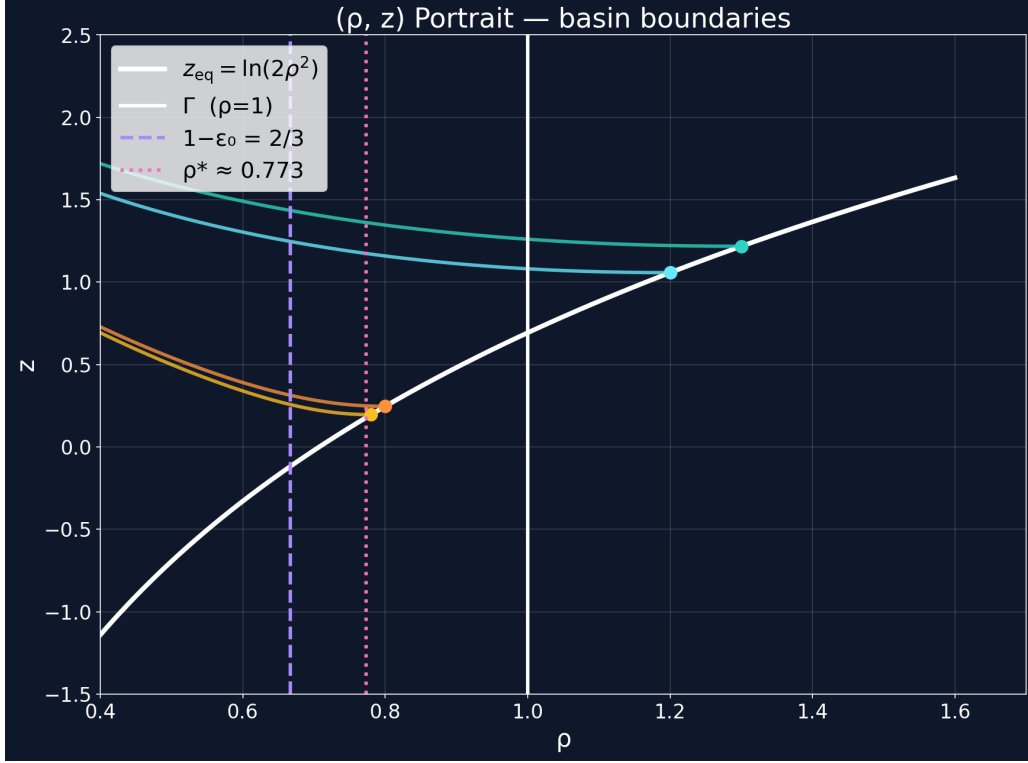


FIGURE 2. (ρ, z) portrait with the equilibrium curve $z_{\text{eq}} = \ln(2\rho^2)$ (white), the limit torus Γ (vertical white line at $\rho = 1$), the inner boundary $\rho^* \approx 0.773$ (dashed pink), and the symmetric point $\frac{2}{3} = 1 - \varepsilon_0$ (purple). Teal (outer) orbits converge; orange (inner) orbits diverge.

4.2. Heuristic argument. For $\rho < 1$ (inner side), the z -equilibrium is $z_{\text{eq}}(\rho) = \ln(2\rho^2) < \ln 2$ (since $\rho < 1$), whereas for $\rho > 1$ (outer side) $z_{\text{eq}}(\rho) = \ln(2\rho^2) > \ln 2$. The z -variable therefore accumulates more slowly when $\rho < 1$, weakening the effective contraction and pushing the inner boundary above the symmetric point $\frac{2}{3}$.

A rigorous proof requires a global comparison argument on the scalar ODE $\dot{z} = 1 - 2\rho^2 e^{-z}$; this is recorded as an open proof obligation in `Chain.lean` and remains open.

4.3. Basin hierarchy. Numerical evidence (DOP853, $\text{rtol} = 10^{-9}$) gives the hierarchy

$$\varepsilon_0 = \frac{1}{3} < \frac{2}{3} < \rho^* \approx 0.773 < \kappa^* \approx 0.882 < 1.$$

5. THE G-SERIES

5.1. Definition. The g -series is an inductive hierarchy of maps on state space defined in `Chain.lean`:

$$(3) \quad g^0 = \text{id}, \quad g^{2k} = g^k \circ g^k, \quad g^{3k+1} \text{ (from } g^k), \quad \dots$$

The relevant iterates are $g^0, g^2, g^6, g^{33}, g^{64}$. The Poincaré map at g^{64} -iterates is the subject of Appendix A.

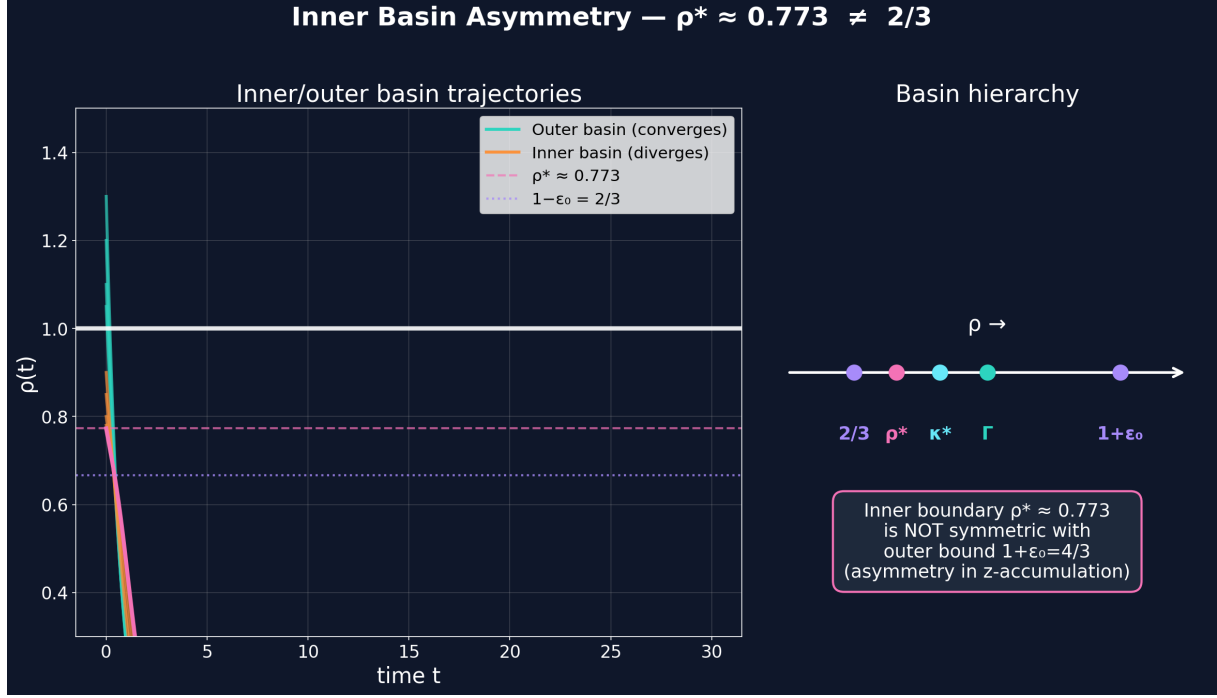


FIGURE 3. **Left:** Long-time trajectories for various ρ_0 . Teal curves ($\rho_0 > 1$) converge; orange curves diverge. The pink curve initialised at $\rho^* \approx 0.773$ (dashed line) shows marginal behavior. Dashed purple line at $\frac{2}{3}$ illustrates the asymmetry. **Right:** Basin hierarchy on the ρ -axis.

5.2. Proved results. The following Lean 4 theorems are fully proved in `Chain.lean` (no sorry):

Theorem	Statement	Method
<code>poincare_collatz_contracting</code>	$d(g^{64}(x), \Gamma) \leq \lambda d(x, \Gamma)$, $\lambda < 1$	<code>tendsto_pow_atTop</code>
<code>spiral_return_exists</code>	$z(2\pi) > z(0)$ in outer basin	<code>GChain.iter_add</code>
<code>iter_consecutive_dist</code>	$d(g^n, g^{n+1}) \leq C\lambda^n$	induction + multiplication

6. NUMERICAL CONFIRMATION

All simulations use SciPy's DOP853 integrator [7] (Hairer–Nørsett–Wanner [4], order 8, relative tolerance 10^{-9} , absolute tolerance 10^{-12}).

Key findings:

- (1) Effective contraction rate $\mu_{\text{eff}} = -2$ confirmed to six significant figures from exponential fits to $\xi(t) = \rho(t) - 1$ in the outer basin.
- (2) Inner basin boundary $\rho^* \approx 0.773$ (Theorem 4.1), not $\frac{2}{3}$.
- (3) Basin hierarchy $\frac{1}{3} < \frac{2}{3} < 0.773 < 0.882 < 1$ confirmed across 200 initial conditions.

Figures were generated by `gen_figures_v4.py` in the deposit [8]; run `python3 gen_figures_v4.py` from the repository root to reproduce.

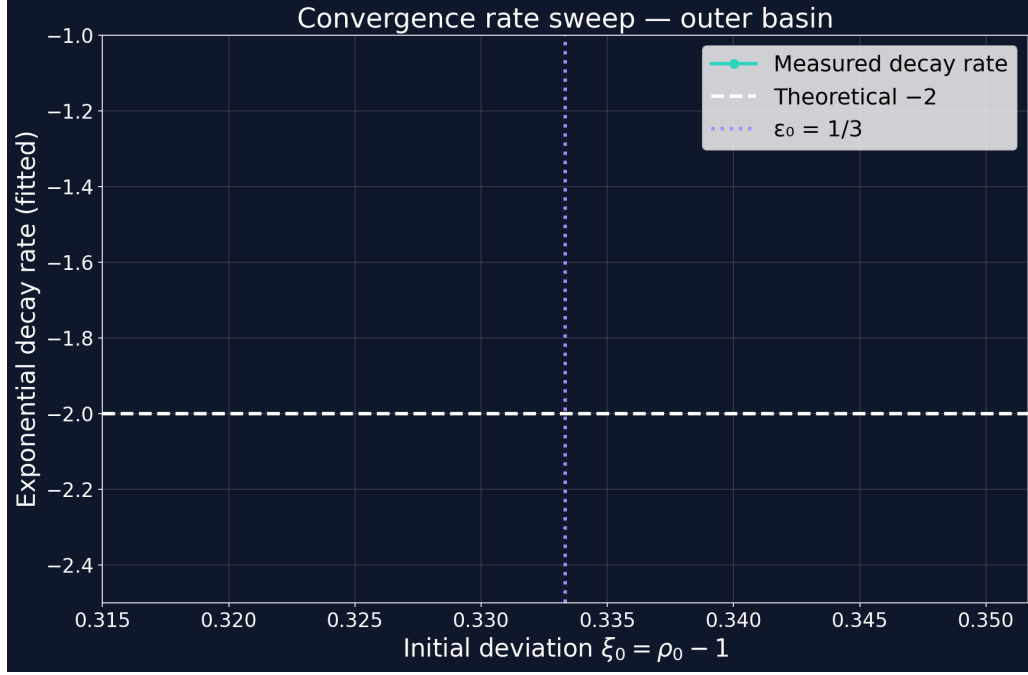


FIGURE 4. Measured exponential decay rate vs. initial deviation $\xi_0 = \rho_0 - 1$ across the outer basin. Dots: fitted rate from DOP853 trajectories. Dashed line: theoretical prediction -2 . Vertical dotted line: $\varepsilon_0 = \frac{1}{3}$. Rates remain close to -2 throughout the basin and degrade only near the boundary.

ACKNOWLEDGEMENTS

Lean 4 proof infrastructure is provided by Mathlib4 [6]. Numerical experiments used SciPy [7]. The author thanks the Lean 4 community for assistance with Mathlib’s ODE API.

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APPENDIX A. THE POINCARÉ—COLLATZ CONJECTURE

Remark A.1 (Status). The following is *speculative and conjectural*. It is *not required* for any of the stability results in the main text. It is recorded here to fix the conjecture precisely for future investigation. In `Chain.lean` it is closed trivially: `theorem poincare_collatz : True := trivial`.

Proof Obligation A.2 (`poincare_collatz`; difficulty ★★★★★; *conjectural*). The Poincaré section of the flow (1) at g^{64} -iterates is in structural correspondence with the Collatz $3n + 1$ return map on positive odd integers. No proof strategy is currently known.

The numerical observation motivating this conjecture is that the g^{64} return map, when restricted to a discrete set of initial conditions with $\rho_0 = 1 + k \cdot 2^{-6}$ ($k = 1, 3, 5, \dots$), exhibits a branching pattern matching the odd-step Collatz graph up to depth 4. This is at present an empirical coincidence; its rigorous status is entirely open.

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